Roll No | 1 | 2 | 3 | 1 | 2 | 0 | 2 | 1 |

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in

B.Tech III Sem I mid semester Examination, 18th Sept, 2024.

Mid Sem-I	Subject: Discrete Mathematics	Course Code: ITC [illegible] 301
Branch: IT [illegible] III Sem	Maximum Marks. -15 marks	Time- 60 min.

Note: All questions are compulsory.

Write every step very carefully.

Q1a)	State the strong induction principle. Using strong induction principle that $(3+\sqrt{5})^n+(3-\sqrt{5})^n$. Is divisible by 2^n $\forall n \geq 0$.	(3 marks)
Q-2	Determine the coefficient of $x^5y^{10}z^5w^5$ in $(x-7y+3z-w)^{25}$ using multinomial theorem.	(3 marks)
Q-3	Let R be the relation on A= {a, b, c, d, e} and *R*= {(a, b), (a, c), (b, c), (c, d), (d, b), (d, e), (e, a) }. Find the transitive closure of *R* by Warshall's algorithm and composition method.	(3 marks)
Q-4	Let R denote relation on the set of ordered pairs of positive integers such that (x, y) *R* (u, v) iff xv=yu. Show that R is an equivalence relation	(3 marks)
Q-5	Find the number of integral solutions to $x_1+x_2+x_3+x_4=50$ where $x_1 \geq -4, x_2 \geq 7, x_3 \geq -14\ x_4 \geq 10$	(3 marks)

$$(3+\sqrt{5})^n + (3-\sqrt{5})^n = 2^n z$$

$$\left(\frac{3+\sqrt{5}}{2}\right)^n + \left(\frac{3-\sqrt{5}}{2}\right)^n = z$$

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Mid Sem-I | **Subject: Design and Analysis Algorithm (UIT302)** | **Roll no** 12312021

Branch: IT | **M.M.-15 marks** | **Time- 60 mint.**

(3rd SEM)

Note: All questions are compulsory.

Marks

Q-1: (a) Define Algorithm. Write a descriptive note on its best case, average case, and worst-case time efficiency. **2+2**

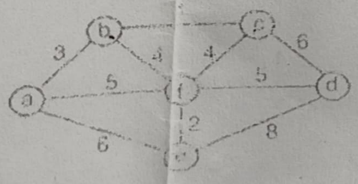
(b) Solve the following recurrence relation using Master's theorem-

$$T(n) = 3T\left(\frac{n}{2}\right) + n^2$$

Q-2: (a) Apply quick sort algorithm to sort the list E, X, A, M, P, L, E in alphabetical order. Draw the tree of recursive calls made. **4**

Q-3: (a) State the Greedy Knapsack. Find the optimal solution to the Knapsack instance n=3, m=20, (P1, P2, P3) = (25, 24, 15) and (W1, W2, W3) = (18, 15, 10). **4**

Q-4: (a) Give the Pseudo code of the Prim's algorithm and apply the same to find the minimum spanning tree of the graph shown below: **3**



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Mid Sem-I | Subject: Object Oriented Programming using JAVA (UIT304) | Roll no ~~12312021~~

Branch: IT (3rd SEM) | Maximum Marks: 15 | Time: 60 minutes

Note: All questions are compulsory.

Marks

Q-1: (a) How many times is the following loop body repeated? What is the output of the loop? — 1

```
int i = 0;
while (i < 10) {
if ((i + 1) % 2 == 0)
System.out.println(i);
i++; }
```

1 3 5 7 9

(b) Convert the following for loop into a do-while loop. — 1

```
int sum = 0;
for (int i = 0; i < 100; i++) { sum += i; }
```

(c) Convert the following if statement using a switch statement — 1

```
// Find interest rate based on year
if (numOfYears == 7)
annualInterestRate = 7.25;
else if (numOfYears == 15)
annualInterestRate = 8.50;
else if (numOfYears == 30)
annualInterestRate = 9.0;
else {
System.out.println("Wrong number of years");
System.exit(0);
}
```

(d) — 1

```
public class Test1 {
public static void main(String[] args) {
int i = 1;
while (i < 10) {
System.out.print(i + " ");
i = i + 3;
}}}
```

1 4 7

Q-2: (a) Write a program that prompts the user to enter an integer n (assume n >= 2) and displays its largest factor other than itself. — 2

(b) Design a class Polynomial that models a polynomial such as 3x^2 + 5x + 2. The class should have: — 2

- A constructor that takes the degree of the polynomial.
- A method addTerm(int coefficient, int power) to add a term to the polynomial.

- A method evaluate(int x) that returns the value of the polynomial for a given x.
- A method toString() to represent the polynomial as a string.

Write a program to demonstrate how you would create and evaluate the polynomial 4x^3 - 2x^2 + 3x + 1.

Q-3: (a)	Explain how Java handles memory management for primitive data types and objects. Include the role of the JVM, stack, and heap memory in your explanation.	2
(b)	Write a Java method findMaxIn2DArray(int[][] array) that takes a two-dimensional array as input and returns the maximum value in the array.	2
Q-4 (a)	Design a class Rectangle that has the following attributes:	3

- width (double)
- height (double)

Include the following methods:

- A constructor that initializes the width and height.
- A method area() that returns the area of the rectangle.
- A method perimeter() that returns the perimeter of the rectangle.

Write a program that demonstrates the use of the class by creating a Rectangle object and printing its area and perimeter.

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Mid Sem-I	Subject: CO(ITC303)	Roll no 12312021
Branch: IT	M.M.-15 marks	Time- 60 mint.

Note: All questions are compulsory.

Q-1	Discuss Flynn's Classification of Computers and provide examples for each category (SIMD, MISD, MIMD).	(3 marks)
Q-2	What is the role of caches in computer architecture? How do they enhance system performance?	(3 marks)
Q-3	Explain different addressing modes and give examples where each mode might be used.	(3 marks)
Q-4	What are the key differences between RISC and CISC processors? Provide advantages and disadvantages of each architecture.	(3 marks)
Q-5	A computer uses a cache memory of size 8 KB. The main memory size is 128 KB with a block size of 64 bytes. Calculate the number of blocks in the main memory and the number of blocks in the cache.	(3 marks)

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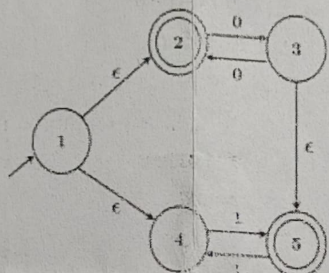
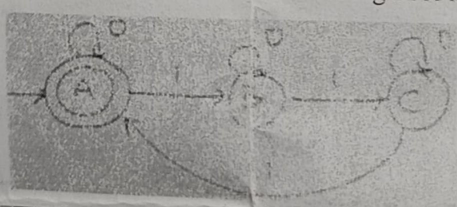
Mid Sem-I	Subject: AFL (UIT-305)	Roll no 12312021
Branch: IT	M.M.-15 marks	Time- 60 mint.

Note: All questions are compulsory.

Q-1	Let M be the 5 state NFA with ε-transitions shown in below figure. Write down regular expressions represents the language accepted by M. 	3.5
Q-2	Design finite automata (FA) machine and corresponding Regular Expression (RE) for the following languages over alphabet ∑(a, b): (a) L start with “ab” & w=2(mod4), where w is a string over ∑(a, b). (b) L accepts strings that do not end with “baa”. (c) L that contains even number of “a” and odd number of “b”. (d) L that contains binary numbers whose equivalent is divisible by 5. (e) L that contains $n_a(w)$ mod 3=0 && $n_b(w)$ mod 4=3	(1.5+15+1.5+1.5+15)=6.5
Q-3	Write down state elimination method and convert following DFA into RE. 	2
Q-4	Let L_1 be the language represented by the regular expression $b^*ab^*(ab^*ab^*)^*$ and $L_2 = \{w \in (a+b)^* \mid \lvert w \rvert \leq 4\}$, where $\lvert w \rvert$ denotes the length of string w. Write down all strings in L2 which is also in L1.	3